

Week 1 - Day 3b: For Loops

You will be working on a series of exercises for the material covered in lecture, including:

- **For Loops**
- **Loop Control Statements**

Lecture Recap: For Loops

A **for loop** is another type of control flow statement used for iterating over a sequence (like a list, tuple, dictionary, set, string, or range). It is less flexible than a **while loop** but is generally easier to read and write for simple iterations.

The syntax for for loops is:

```
for <variable> in <iterable expression>:  
    <expression> / <body>
```

`range()` allows you to loop a specific number of times or access sequence elements by index.

The syntax for `range()` is:

- `range(n)` creates a sequence from 0 to $n - 1$
- `range(start, stop)` creates a sequence from `start` to `stop - 1`
- `range(start, stop, step)` lets you specify a step size

Exercise 1: Iterative Printing of Numbers []

1.1 Print the integers from 0 - 10 using a `for` loop.

```
# TODO: Your code here
```

1.2 Print the integers from 0 - 10 using a `while` loop

```
# TODO: Your code here
```

1.3 Print the integers from 0 - 100 that are divisible by 7 in decreasing order using a `for` loop.

```
# TODO: Your code here
```

1.4 Print the integers from 0 - 100 that are divisible by 7 in decreasing order using a `while` loop.

```
# TODO: Your code here
```

1.5 Print the integers from 0 - 100 that are divisible by 7 in decreasing order using a `for` loop, but without using an `if` statement.

```
# TODO: Your code here
```

Exercise 2: More Loops []

2.1 Write a `while` loop that prints the integers between 0 and 10 (inclusive on both ends).

```
# TODO: Your code here
```

2.2 Change the loop above to be a `for` loop.

```
# TODO: Your code here
```

Exercise 3: Enumerating a List []

3.1 Let `lst = [12, 34, 15, 11, 9, 34, 433, 23, 24, 121, 343, 32, 111, 4454, 323]` be a list of integers. Use a `while` loop to print all the elements in `lst`

```
# TODO: Your code here
```

```
lst = [12, 34, 15, 11, 9, 34, 433, 23, 24, 121, 343, 32, 111, 4454, 323]
```

3.2 Change the loop above to be a `for` loop.

```
# TODO: Your code here
```

3.3 Print only the odd elements (e.g. 15, 11, 9, ...) in `lst` using a `for` loop.

```
# TODO: Your code here
```

3.4 Print every other element in `lst` using a `for` loop.

```
# TODO: Your code here
```

Lecture Recap: Loop Control Statements

You can control the flow of a loop using special keywords:

`continue` skips the rest of the current iteration and jumps to the next one.

`break` exits the loop entirely when a condition is met.

Exercise 4: Loop Control Statements []

4.1 Write code that prints numbers from 0 to 20, excluding multiples of 3, utilizing the `continue` statement.

TODO: Write your code here

4.2 Try running the code below and see what happens (hint: you might need to use the stop button at the top of the page).

```
i = 0
while True:
    i += 1
    print(i)
```

4.3 Now, edit the code above to only print the numbers up to 20, using `break`.

TODO: Copy in the code from 4.2 and edit it.

4.4 Now, using `continue`, use the same code from 4.2 to print the numbers up to 20 that are not multiples of 3.

TODO: Copy in the code from 4.2 and edit it.

Congratulations 🎉